

xiexiang_POS_oral_exam_review_from_PPT 1

POS oral exam review notes - answers from teacher PPT

中文：POS口试复习笔记-老师解答PPT

Source rule: answers below are based on the course PPT PDFs in ppt/ . I keep the teacher's wording as much as possible and add source tags like [Pos09 p6] . Formulas are rewritten in clean notation when the PDF text extraction corrupted symbols, but the meaning and parameters follow the PPT pages.

中文：来源规则：以下答案基于 ppt/ 中的课程PPT PDF。我尽可能保留老师的措辞，并添加源标签，如 [Pos09 p6] 。当 PDF 文本提取损坏符号时，公式会以干净的符号重写，但含义和参数遵循 PPT 页面。

Exam strategy:

中文：考试策略：

- First answer with the short PPT definition.
- If the teacher asks "details?", add formula, parameters, and error budget.
- If the teacher expands randomly, use the "Possible expansion" bullets.

1. First level: answer or leave

1.1 General concepts from Introduction

Equation of the distance between 2 points

For two points $P_1 = (X_1, Y_1, Z_1)$, $P_2 = (X_2, Y_2, Z_2)$:

$$d(P_1, P_2) = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2 + (Z_2 - Z_1)^2}$$

In positioning, if the signal travel time is T , the distance is converted as:

$$d = cT$$

where c is the propagation speed of the electromagnetic signal and T is the travel time.

The PPT describes that receivers observe signal travel times, convert them to distances, and use the observed distances to estimate the receiver position. [Pos01 p23, p32]

中文：其中 c 是电磁信号的传播速度， T 是传播时间。PPT描述了接收机观察信号传播时间，将其转换为距离，并使用观察到的距离来估计接收机位置。 [Pos01 p23, p32]

Possible expansion:

中文：可能的扩展：

- With 1 distance from 1 known point: the receiver lies on a sphere/circle around the known point.
 - 中文：与 1 个已知点的距离为 1：接收机位于已知点周围的球体/圆上。
- With 2 distances from 2 known points: intersection reduces possible positions.
 - 中文：与 2 个已知点的距离为 2：相交减少可能的位置。
- With 3 distances from 3 known points: position can be determined geometrically. [Pos01 p20–p22]
 - 中文：距离 3 个已知点有 3 个距离：可以通过几何方式确定位置。 [Pos01 p20–p22]

Definition of Accuracy

Accuracy is the quality of positioning, defined as the error of the estimated position with respect to the true position.

It is usually described separately for horizontal水平 and vertical垂直 components, It is usually given with a confidence level, typically 95%. [Pos01 p5]

Useful examples from PPT:

- Citizen outdoor navigation: about 10 m at 0.1 Hz.
- Fast static surveying: better than 10 cm.
- Long static monitoring: from 1 cm to 1 mm. [Pos01 p16]

Possible expansion:

中文：可能的扩展：

- Besides accuracy, positioning systems are also described by latency, update frequency, availability, coverage, cost, and user requirements. [Pos01 p14, p481–p512]
 - 中文：除了准确性之外，定位系统还可以通过延迟、更新频率、可用性、覆盖范围、成本 and 用户需求来描述。 [Pos01 p14, p481–p512]

Definition of kinematic and static positioning

Kinematic positioning:

the receiver is moving, so the coordinates are time-dependent and estimated epoch by epoch, typically at 1 Hz [Pos01 p15, p18]

Static positioning:

the receiver is assumed to be stationary during the observation session, so the coordinates are constant in time and are estimated from multiple epochs.

Static positioning is mainly used in monitoring and surveying;

[Pos01 p15, p16]

中文：静态定位：接收机固定；一个位置是通过一段时间内的观察来估计的。 [Pos01 p15, p16]

Definition of real time and post processing results

Real time positioning:

the position solution is produced immediately, or with very low latency, it is mainly required for navigation, where positioning errors may directly lead to unsafe or incorrect actions. [Pos01 p17]

Post processing:

data are acquired and stored in real time, then processed later in the office. It is common for static surveying and monitoring and can reach higher accuracy. [Pos01 p17]

Definition of outdoor / indoor positioning

Outdoor positioning: open sky environment. GNSS can satisfy many accuracy and latency requirements, from citizen navigation with mass-market devices to real-time surveying with professional instrumentation. [Pos01 p28]

Indoor positioning: positioning within buildings such as hospitals, shopping malls, industries, and homes. GNSS is not available or signals are badly corrupted, so other infrastructure and technologies are needed. [Pos01 p29, Pos13 p1]

Examples of user requirements

Users requirements mainly include accuracy, latency, temporal resolution, integrity, availability, coverage area and so on.

1. Accuracy Accuracy describes how close the estimated position is to the true position, usually expressed at a given confidence level (e.g. 95%).
2. Latency Latency is the time needed to produce a position estimate, ranging from real-time requirements to cases where delay is not critical.
3. Temporal resolution Temporal resolution refers to how frequently position updates are provided, from high-rate (e.g. 10 Hz) to very low-rate (monthly or yearly).
4. Real-time vs post-processing Real-time and post-processing modes are chosen according to latency and temporal resolution requirements of the application.
5. Availability Availability is the percentage of time during which a positioning solution is successfully provided.
6. Coverage area Coverage area defines the spatial extent where a positioning technology works, from local indoor spaces to global scale.
7. Needed output A positioning system may output not only position, but also time, velocity, acceleration, heading, or direction.
8. Output position type Position output can be expressed in different forms, such as full 3D coordinates, 2D floor-level positions, or proximity information.

9. Integrity Integrity is the system's ability to detect and warn the user when the positioning error may exceed a predefined alert limit within a specified time. For example, in mass-market LBS like car navigation, it needs low latency and high update rate bc the position must refresh quickly, global coverage, and meter-level accuracy is usually enough. And for Geodetic monitoring / reference networks, it needs very high accuracy (cm to mm) and high stability, but latency is less important because results can be computed in post processing (daily/weekly solutions).
For Ambient Assisted Living, providing positioning and monitoring for elderly or vulnerable people. It needs high availability at home, room-level coverage, and usually meter-level accuracy is enough. Latency requirements depend on the function: for safety-critical events (e.g., fall detection) we need low latency, typically seconds.

1.2 Reference Frames

Definition of reference systems in 1, 2 and 3 dimensions

****Global reference systems**** cover the Earth and are realized by global networks of permanent stations.

Local reference systems are defined at continental, national, or very local scale for surveying, mapping, monitoring, and positioning. `[Pos03 p25-p26]`

中文：全球参考系统覆盖地球，并由全球永久站网络实现。本地参考系统是在大陆、国家或当地范围内定义的，用于测量、测绘、监测和定位。 [Pos03 p25-p26]

A 1D reference system is composed of:

1. a unit of length (scale);
2. an origin (zero point).

Then the coordinate X_p is simply a signed distance of P from the origin along the line. If I change the origin, the coordinate of the same point changes by a translation between the two origins P_{zero} and $P_{\text{zero_prime}}$.

In 2D, a reference system is defined by:

- two orthogonal axes (typically East and North),
- one unit of lengths,
- and an origin.

The coordinates of a point P are the orthogonal projections of the position vector on the two axes, i.e. (E_P, N_P)

P is the point,

$\vec{p}$ is the vector between the origin and P ,

$$\mathbf{p} = \begin{bmatrix} E_P \\ N_P \end{bmatrix}$$

is P position wrt the axes.

In 2D, the coordinates of the same point transform between two reference systems by a translation and a rotation (and, if needed, a scale factor):

$$1, \mathbf{p}'' = \mathbf{t} + \lambda \mathbf{R} \mathbf{p}', \quad 2, \mathbf{p}' = \lambda^{-1} \mathbf{R}^T (\mathbf{p}'' - \mathbf{t})$$

1, It is the forward conversion formula that converts the coordinate $\mathbf{p}'$ in the local coordinate system to the coordinate $\mathbf{p}''$ in the target coordinate system.

是正向转换公式，将局部坐标系下的坐标 $\mathbf{p}'$ 转换到目标坐标系下的坐标 $\mathbf{p}''$

2, It is a forward conversion formula that restores the coordinates in the **target coordinate system back to** those in the **source coordinate system**.

是正向转换公式，将目标坐标系坐标还原回源坐标系坐标。

$|\mathbf{p}'|$ 源坐标系下的点位坐标向量

$|\mathbf{p}''|$ 目标坐标系下转换后的点位坐标向量

$|\mathbf{t}|$ 2D translation vector 二维平移向量（坐标原点偏移量） $\mathbf{t}=[t_E, t_N]$ 转置

$|\lambda|$ 2D translation vector 尺度缩放系数（两个坐标系单位长度的比例，无缩放时 $\lambda=1$ ）

$|\mathbf{R}|$ Two-dimensional orthogonal rotation matrix 二维正交旋转矩阵， $\mathbf{R}=[\text{旋转矩阵}]$ ， θ 为坐标轴旋转角

In 3D, a reference system is defined by:

- three orthogonal axes with a common origin,
- and a unit of lengths (unit vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$).

The coordinates of a point are the lengths of its orthogonal projections on the three axes (Cartesian coordinates)

Definition of ITRS

ITRS is the International Terrestrial Reference System.

- It is geocentric, Earth-fixed right-handed Cartesian reference system.
 - Unit of length: meter.
 - It is consistent with the TCG scale.
 - 中文：与TCG量表一致。
 - Its realization is the ITRF. [Pos03 p29]
 - 意思是ITRS：是一套「理论上的参考规则体系」，ITRF：是这套理论规则「实际落地的坐标数据集」。 [Pos03 p29]
- define ITRS and explain its relationship with ITRF.
- Definition of ITRS

ITRS stands for the International Terrestrial Reference System.

It is a geocentric, Earth-fixed right-handed Cartesian reference system. ITRS is a theoretical abstract standard with strict mathematical definitions. Its practical realization is called ITRF.

Global Cartesian coordinates of a point

A point is represented by geocentric Cartesian coordinates:

$$\mathbf{x}_P = [X_P, Y_P, Z_P]^T$$

In geodetic transformations, the global Cartesian system is connected to geodetic coordinates (φ, λ, h) through the reference ellipsoid. [Pos03 p57-p58]

中文：在大地测量变换中，全局直角坐标系通过参考椭球体连接到大地坐标 (φ, λ, h) 。 [Pos03 p57-p58]

Reference ellipsoid

A reference ellipsoid is a first-order approximation of the Earth's shape, so it is a mathematically defined, smooth surface.

The geoid is irregular and cannot be modeled with simple analytical functions. [Pos03 p51]

中文：大地水准面是不规则的，无法使用简单的分析函数进行建模。 [Pos03 p51]

A rotational ellipsoid is defined by:

中文：旋转椭球定义为：

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{b^2} = 1$$

where:

- a : major semi-axis.
 - 中文： a ：长半轴。
- b : minor semi-axis.
 - 中文： b ：短半轴。

Other parameters:

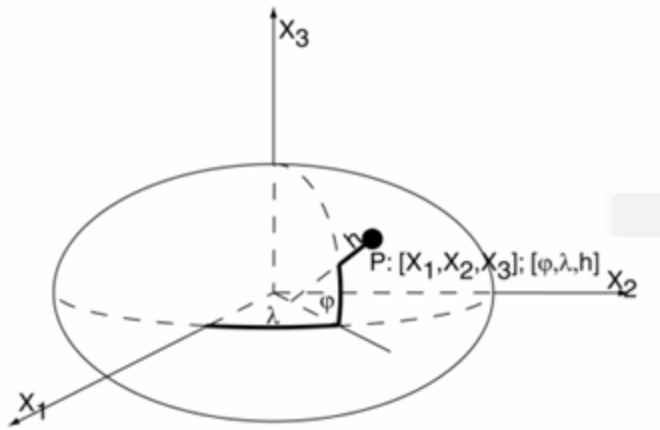
$$f = \frac{a - b}{a}, \quad e^2 = \frac{a^2 - b^2}{a^2}$$

The deviations between the current GRS80 ellipsoid and the geoid have zero mean and maximum values of about 100 m.当前GRS80椭球与大地水准面之间的偏差均值为零，最大值约100米。

Geodetic coordinates

Given a reference frame and the related ellipsoid, the geodetic coordinates of point P are (φ, λ, h) :

- φ , geodetic latitude: angle between the normal to the ellipsoid passing through P and the equatorial plane.
 - 中文: φ , 大地纬度: 穿过 P 的椭球法线与赤道面之间的角度。
- λ , geodetic longitude: angle between the meridian plane passing through P and the origin meridian plane.
 - 中文: λ , 大地经度: 经过 P 的子午面与原点子午面之间的角度。
- h , geodetic height: distance between P and the ellipsoid surface. [Pos03 p56]
 - 中文: h , 大地高: P 与椭球面之间的距离。 [Pos03 p56]



Transformation from geodetic to Cartesian:

中文: 从大地测量到笛卡尔坐标的转换:

$$X_P = (R_N + h_P) \cos \varphi_P \cos \lambda_P$$

$$Y_P = (R_N + h_P) \cos \varphi_P \sin \lambda_P$$

$$Z_P = [R_N(1 - e^2) + h_P] \sin \varphi_P$$

$$R_N = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi_P}}$$

Parameters:

- X_P, Y_P, Z_P : global Cartesian coordinates of P .
 - 中文: X_P, Y_P, Z_P : P 的全局直角坐标坐标。
- $\varphi_P, \lambda_P, h_P$: geodetic latitude, longitude, geodetic height.
 - 中文: $\varphi_P, \lambda_P, h_P$: 大地纬度、经度、高程。
- R_N : grand normal radius, East-West curvature radius.
 - 中文: R_N : 大法线半径, 东西曲率半径。

- a : ellipsoid major semi-axis.
 - 中文: a : 椭球长半轴。
- e : ellipsoid eccentricity /ɪˈlɪpsɔɪd ˌɛksənˈtrɪsəti/. [Pos03 p57]
 - 中文: e : 椭球偏心率。 [Pos03 p57]

Possible expansion: teacher may ask "equator/latitude/longitude definition".

赤道定义: The line of intersection between the equatorial plane and the Earth's surface

- Equatorial plane: plane used as reference for geodetic latitude.
 - 中文: 赤道面: 用作大地纬度参考的平面。
- Geodetic latitude is measured from the ellipsoid normal to the equatorial plane.
 - 中文: 大地纬度是从垂直于赤道面的椭球体测量的。
- Geodetic longitude is measured between the point meridian plane and the origin meridian plane. [Pos03 p56]
 - 中文: 大地经度是在点子午面和原点子午面之间测量的。 [Pos03 p56]

Local Cartesian reference system

Local Cartesian coordinates are East, North, Up coordinates. Given an origin point for the interest area, nearby points are expressed as East, North, Up with respect to the origin. [Pos03 p58, p60]

中文: 局部直角坐标是东、北、上坐标。给定兴趣区域的原点, 附近的点相对于原点表示为东、北、上。 [Pos03 p58, p60]

The Local Cartesian reference system is a local 3D frame defined at a point P_0 on the ellipsoid. Use:

中文: 使用:

- Navigation in local scenarios.
 - 中文: 本地场景导航。
- Local deformation monitoring.
 - 中文: 局部变形监测。
- Engineering/surveying applications. [Pos03 p60–p62]
 - 中文: 工程/测量应用。 [Pos03 p60–p62]

Body frame reference system

The body frame is a local 3D reference system attached to the vehicle, so it moves and rotates together with the body. A typical definition:

中文: 车身框架用于导航。一个典型的定义:

- Origin in a generic point inside the body, for example the barycenter.
- X : main vehicle axis, motion direction.

- Y : perpendicular to X in the vehicle plane.
- Z : perpendicular to the vehicle plane. [Pos03 p66]

Euler angles :

It is used to indicate the deflection angle of the body relative to the Geodetic coordinate system, with the three angles corresponding to rotations around the three respective axes.

- roll α : rotation around X .
 - 中文：滚动 α ：围绕 X 旋转。
- pitch β : rotation around Y .
 - 中文：俯仰 β ：围绕 Y 旋转。
- yaw γ : rotation around Z . [Pos03 p67]
 - 中文：偏航 γ ：围绕 Z 旋转。 [Pos03 p67]

Transformation:

$$\mathbf{x}_{LC} = R_Z(\gamma)R_Y(\beta)R_X(\alpha)\mathbf{x}_{BF}$$

Important PPT warning: different definitions of Y , Z , roll, pitch, yaw imply different rotation angles and concatenations, so definitions must be checked. [Pos03 p68]

重要 PPT 警告： Y , Z 、横滚、俯仰、偏航的不同定义意味着不同的旋转角度和串联，因此必须检查定义。 [Pos03 p68]

Definition of geoid

中文：大地水准面的定义

The geoid is the equipotential surface of the gravity field that passes through the mean sea level. It is an irregular surface, smooth but not representable by simple analytical functions.

[Pos03 p51]

中文：大地水准面是穿过平均海平面的重力场等位面。它是一个不规则的表面，光滑但不能用简单的解析函数表示。 [Pos03 p51]

Geoid undulation, orthometric height and height transformations

中文：大地水准面起伏、正高和高程变换

Geoid undulation $N(\varphi, \lambda)$ is the height of the geoid with respect to the reference ellipsoid.

[Pos03 p54]

中文：大地水准面起伏 $N(\varphi, \lambda)$ 是大地水准面相对于参考椭球的高程。 [Pos03 p54]

GNSS gives ellipsoidal heights h , but engineering and environmental applications usually need orthometric heights H , because they represent physical quantities. [Pos03 p74]

中文：GNSS 给出椭球高程 h ，但工程和环境应用通常需要正高 H ，因为它们代表物理量。
[Pos03 p74]

Height conversion:

中文：身高换算：

$$h = H + N$$

or:

$$H = h - N$$

Parameters:

中文：参数：

- h : ellipsoidal height from GNSS.
 - 中文： h ：来自 GNSS 的椭球高程。
- H : orthometric height.
 - 中文： H ：正交高程。
- N : geoid undulation.
 - 中文： N ：大地水准面波动。

Height difference conversion:

中文：身高差换算：

$$\Delta H = \Delta h - \Delta N$$

The PPT stresses that this conversion can never be neglected. [Pos03 p75]

中文：PPT强调这种转变绝对不能忽视。 [Pos03 p75]

1.3 GNSS

中文：1.3 全球导航卫星系统

General description of GPS, GLONASS, Beidou and Galileo

中文：GPS、GLONASS、北斗和伽利略的一般描述

GNSS provides free navigation data and observables for users; main observables are pseudoranges and phase observations. [Pos05 p2]

中文：GNSS为用户提供免费的导航数据和观测数据；主要观测值是伪距和相位观测值。
[Pos05 p2]

PPT summary: in the past positioning was based on GPS; now it uses GPS + Galileo + BeiDou + GLONASS. [Pos05 p10]

Possible expansion: **differences between constellations.**

中文：可能的扩展：星座之间的差异。

- GPS and GLONASS signal access: GPS is based on codes (CDMA), GLONASS historically on phases/frequencies (FDMA).
- GPS, Galileo, BeiDou provide several frequencies and codes.
- GPS transmits Klobuchar ionospheric model, GLONASS does not transmit an ionospheric model, BeiDou transmits BDGIM, Galileo transmits NeQuick. [Pos07 p16–p20, Pos08 p9]
 - 中文：GPS传输Klobuchar电离层模型，GLONASS不传输电离层模型，北斗传输BDGIM，伽利略传输NeQuick。 [Pos07 p16–p20, Pos08 p9]

General description of s and phase carriers

GNSS signals contain:

中文：GNSS 信号包含：

1. sinusoidal carrier phases, in GPS named L1, L2, L5;
2. binary codes, in GPS named C/A, P(Y), L2C, L5, L1C, M;
3. navigation message, in GPS named D. [Pos07 p1–p2]

Binary codes are pseudo-random sequences of chips, typically taking values +1 and −1. The code observation, or pseudorange, is obtained by comparing the received code with the receiver internal replica. [Pos07 p9, p31–p40]

中文：二进制码：等于+1和−1的脉冲序列。通过将接收到的代码与接收机内部副本进行比较来获得代码观测值或伪距。 [Pos07 p9, p31–p40]

Carrier phase: a sinusoidal oscillator signal modulated by binary codes. Phase observations are used for high precision positioning because electronic noise is about 1-2 mm, but they contain ambiguity. [Pos07 p10–p12, Pos10 p9]

中文：载波相位：由二进制码调制的正弦振荡信号。相位观测用于高精度定位，因为电子噪声约为 1-2 毫米，但它们包含模糊度。 [Pos07 p10–p12, Pos10 p9]

Introduction to atmospheric effects

Atmospheric effects include:

- ionospheric delay;
 - 中文：电离层延迟；
- tropospheric delay;
 - 中文：对流层延迟；
- multipath. [Pos08 p1]

The ionosphere is dispersive: the effect depends on carrier frequency. The sign is positive for codes and negative for carrier phases. [Pos08 p5-p7]

中文：电离层是色散的：其影响取决于载波频率。代码的符号为正，载波相位的符号为负。
[Pos08 p5-p7]

The troposphere depends on air and water vapor content from 0 to 40 km height. It is not frequency dependent for GNSS in the same way as the ionosphere. [Pos08 p14-p16]

中文：对流层取决于 0 至 40 公里高程的空气和水蒸气含量。它不像电离层那样依赖于 GNSS 的频率。 [Pos08 p14-p16]

Description of multipath

#Multipath is typical in urban scenarios. It is due to reflection of the GNSS signal when approaching the receiver antenna. Its mathematical modeling is complicated and not intuitive. It can reach:

中文：多路径在城市场景中很常见。这是由于 GNSS 信号接近接收天线时发生反射。其数学模型复杂且不直观。它可以达到：

- some centimeters for phases;
 - 中文：几厘米的相位；
- several meters for codes;
 - 中文：代码数米；
- periodic behavior, with a period of about 20 minutes. [Pos08 p21]
 - 中文：周期性行为，周期约20分钟。 [Pos08 p21]

For SPP, multipath can be a major part of the error budget and severe multipath can degrade both positioning and quality checks. [Pos09 p22-p24]

中文：对于 SPP，多路径可能是错误预算的主要部分，严重的多路径会降低定位和质量检查的性能。 [Pos09 p22-p24]

For PPP phase observations, multipath is not the direct sum of the excess path and does not exceed a few cm in the worst cases. [Pos10 p17]

中文：对于 PPP 相位观测，多径不是多余路径的直接和，并且在最坏的情况下不会超过几厘米。 [Pos10 p17]

Possible expansion: "How many satellites are needed to remove errors?"

中文：可能的扩展：“需要多少颗卫星才能消除错误？”

- SPP solves 4 unknowns in the simplest case: 3 receiver coordinates plus receiver clock offset, so at least 4 satellites are needed for a single epoch solution.
 - 中文：SPP 在最简单的情况下求解 4 个未知数：3 个接收机坐标加上接收机钟差，因此单个历元解至少需要 4 个卫星。

- More satellites give redundancy for Least Squares and quality control, but they do not automatically eliminate all systematic errors.
 - 中文：更多卫星为最小二乘法和质量控制提供冗余，但它们不会自动消除所有系统误差。
- Dual-frequency observations can remove first-order ionospheric delay through ionosphere-free combination.
 - 中文：双频观测可以通过无电离层组合消除一阶电离层延迟。
- Difference techniques can cancel or reduce receiver/satellite clock errors and common atmospheric/orbit errors, depending on geometry and baseline length. [Pos09 p10–p13, Pos09 p24, Pos11 p1–p5]
 - 中文：差分技术可以消除或减少接收机/卫星时钟误差和常见的大气/轨道误差，具体取决于几何形状和基线长度。 [Pos09 p10–p13, Pos09 p24, Pos11 p1–p5]

Standard Point Positioning: observations, unknowns, biases

simplest case:

- code observations;
- single epoch;
- real time;
- single constellation;
- single frequency;
- one code observation for each satellite;
- m satellites in view gives m code observations. [Pos09 p6]
 - 中文：视野中的 m 卫星提供 m 代码观测值。 [Pos09 p6]

Unknowns:

- receiver coordinates X_R, Y_R, Z_R ;
- receiver clock offset dt_R .

So SPP needs at least four satellites in the simplest case. [Pos09 p10–p13]

中文：所以SPP在最简单的情况下至少需要四颗卫星。 [Pos09 p10–p13]

Clean code observation equation:

$$P_{R,i}^{S,G}(t) = \rho_R^S(t) + c[dt_R(t) + d_{R,i}^G - dt^S(t) - d_i^{S,G}] + T_R^S(t) + I_{R,i}^S(t) + \varepsilon$$

Parameters:

- $P_{R,i}^{S,G}$: code/pseudorange observation from receiver R to satellite S , constellation G , frequency i .

- 中文: $P_{R,i}^{S,G}$: 从接收机 R 到卫星 S 、星座 G 、频率 i 的代码/伪距观测。
- ρ_R^S : geometric distance between receiver and satellite.
 - 中文: ρ_R^S : 接收机与卫星之间的几何距离。
- c : speed of light.
 - 中文: c : 光速。
- dt_R : receiver clock offset.
 - 中文: dt_R : 接收机钟差。
- $d_{R,i}^G$: receiver hardware bias for constellation/frequency.
 - 中文: $d_{R,i}^G$: 星座/频率的接收机硬件偏差。
- dt^S : satellite clock offset.
 - 中文: dt^S : 卫星钟差。
- $d_i^{S,G}$: satellite hardware bias.
 - 中文: $d_i^{S,G}$: 卫星硬件偏差。
- T_R^S : tropospheric delay.
 - 中文: T_R^S : 对流层延迟。
- $I_{R,i}^S$: ionospheric delay.
 - 中文: $I_{R,i}^S$: 电离层延迟。
- ε : residual modeling error and measurement noise. [Pos09 p2]
 - 中文: ε : 残余建模误差和测量噪声。 [Pos09 p2]

In simplest single-frequency single-constellation SPP, the receiver hardware bias can be grouped with receiver clock offset into one receiver "clock" parameter. Ionospheric and tropospheric effects are modeled by standard models. [Pos09 p7]

中文: 在最简单的单频单星座 SPP 中, 接收机硬件偏差可以与接收机钟差一起分组为一个接收机“时钟”参数。电离层和对流层效应通过标准模型进行建模。 [Pos09 p7]

SPP biases/error sources:

中文: SPP 偏差/误差源:

1. satellite ephemerides, clock offset and hardware biases;
中文: 1. 卫星星历、钟差和硬件偏差;
2. ionospheric residual;
中文: 2. 电离层残余;
3. tropospheric residual;
中文: 3. 对流层残余;
4. multipath;
中文: 4. 多路径;
5. code measurement noise / data quality. [Pos09 p21-p24]
中文: 5. 代码测量噪声/数据质量。 [Pos09 p21-p24]

Final error budget for SPP

From PPT:

- Broadcast ephemerides/clocks: about 1-2 m; high precision real-time ephemerides/clocks: less than 1 m.
 - 中文：广播星历/时钟：约1-2m；高精度实时星历/时钟：小于1 m。
- Single-frequency ionospheric residual error can reach 20 m.
 - 中文：单频电离层残差可达20 m。
- Tropospheric error is almost always smaller than 20 cm.
 - 中文：对流层误差几乎总是小于 20 厘米。
- Code observation noise: 10-20 cm for new codes/new receivers; 100-200 cm for old GPS codes/old receivers.
 - 中文：码观测噪声：新代码/新接收机为 10-20 cm；旧 GPS 代码/旧接收机为 100-200 厘米。
- Multipath can be severe, several meters for codes, and in bad cases much larger in urban conditions. [Pos07 p41-p43, Pos09 p21-p24, Pos08 p21]
 - 中文：多路径可能很严重，代码可达数米，在城市条件下情况恶劣时会更大。 [Pos07 p41-p43, Pos09 p21-p24, Pos08 p21]



image

[[Pasted

Precise Point Positioning: observations, unknowns, biases

PPP is undifferenced processing of phase observations, like SPP on codes, but with precise modeling and precise products. [Pos10 p10]

中文: PPP 是对相位观测的无差别处理, 类似于代码上的 SPP, 但具有精确的建模和精确的产品。 [Pos10 p10]

PPP is based on multi-frequency phase observations and uses ionosphere-free combinations. [Pos10 p14]

中文: PPP 基于多频率相位观测并使用无电离层组合。 [Pos10 p14]

PPP relies on a posteriori precise satellite navigation data:

中文: PPP 依赖于后验精确的卫星导航数据:

- ephemerides;
- 中文: 星历表;

- clock offsets;
 - 中文：钟差；
- hardware biases.
 - 中文：硬件偏差。

Needed accuracies are at the level of a few centimeters. [Pos10 p15]

中文：所需的精度为几厘米级别。 [Pos10 p15]

Unknowns in single epoch with m satellites and n constellations:

中文：具有 m 卫星和 n 星座的单个历元中的未知数：

1. 3 coordinates;

中文：1. 3个坐标；
2. n receiver clock offsets plus constellation hardware biases;

中文：2. n 接收机钟差加上星座硬件偏差；
3. m ambiguities;

中文：3. m 含糊之处；
4. at least one zenith tropospheric residual.

中文：4. 至少一个天顶对流层残差。

So PPP is not solvable in single epoch; batch processing of multiple epochs is strictly needed, for static and also for kinematic. [Pos10 p18]

中文：所以 PPP 在单个 epoch 中是不可解的；对于静态和运动来说，严格需要多个纪元的批处理。 [Pos10 p18]

Final error budget of PPP

For long static post-processing PPP:

中文：对于长静态后处理 PPP：

- no ionospheric effect by ionosphere-free combination;
 - 中文：无电离层组合不会产生电离层效应；
- tropospheric effect is estimated, not only modeled;
 - 中文：对流层效应是估算的，而不仅仅是建模；
- multipath is handled by accurate outlier rejection;
 - 中文：通过准确的异常值拒绝来处理多径；
- final static PPP can reach centimeter-level accuracy. [Pos10 p24-p25]
 - 中文：最终静态PPP可以达到厘米级精度。 [Pos10 p24-p25]

Main residuals:

中文：主要残差：

- precise product errors at few cm level;
 - 中文：产品误差精确到厘米级；
- residual troposphere estimation;
 - 中文：残余对流层估计；
- phase multipath, usually within a few cm;
 - 中文：相位多径，通常在几厘米内；
- cycle slips / new ambiguities if signal lock is lost. [Pos10 p15-p17, p30]
 - 中文：如果信号锁定丢失，则会出现周跳/新的歧义。 [Pos10 p15-p17, p30]

1.4 INS——Inertial Navigation System: components

INS consists of:

- 3 accelerometers;;
- 3 gyroscopes;
- hardware for collecting data;
- software for real-time processing. [Pos12 p6]

Each sensor component belongs to the IMU, (Inertial Measurement Unit). [Pos12 p6]

中文：每个传感器组件都属于 IMU（惯性测量单元）。 [Pos12 p6]

Inertial Measurement Units and output

中文：惯性测量单位和输出

The IMU has:

中文：IMU 具有：

- 3 accelerometers in three orthogonal directions, fixed in the body frame;
 - 中文：三个正交方向的3个加速度计，固定在车身框架上；
- 3 gyroscopes in three orthogonal directions.
 - 中文：三个正交方向的 3 个陀螺仪。

Outputs:

中文：输出：

- linear/specific accelerations in the body frame;
- angular rates of the sensor/body rotation.

By integration and navigation equations, INS estimates position, velocity and attitude. [Pos01 p34, Pos12 p6-p7, p18]

中文：通过积分和导航方程，INS 估计位置、速度和姿态。 [Pos01 p34, Pos12 p6-p7, p18]

Accelerometer definition:

- Basic principle: accelerometers measure the forces acting on a proof mass.
 - 中文：基本原理：加速度计测量作用在检测质量上的力。
- Open loop accelerometers: measure the displacement of the proof mass caused by external forces.
 - 中文：开环加速度计测量由外力引起的检测质量的位移。
- Closed loop accelerometers: keep the proof mass in balanced by generating an opposite force. [Pos12 p7]
 - 中文：闭环加速度计通过产生相反的力使检验质量保持平衡。 [Pos12 p7]

Gyroscope definition:

- Gyroscopes measure the angular rate of the sensor rotation with respect to an inertial reference system.
 - 中文：陀螺仪测量传感器相对于惯性参考系统旋转的角速率。
- Main types: mechanical and optical.
 - 中文：主要类型：机械式和光学式。
- Fiber optic gyroscopes are based on the Sagnac effect. [Pos12 p12-p13]
 - 中文：光纤陀螺仪基于萨尼亚克效应。 [Pos12 p12-p13]

1.5 Indoor positioning

Applications and user requirements

Indoor positioning has no GNSS but infrastructure and is challenging. Applications include navigation, safety information, robotics, hospitals, security, industries, museums, logistics, parking navigation, surveying and monitoring of buildings. [Pos13 p1-p4]

中文：室内定位没有 GNSS，只有基础设施，具有挑战性。应用包括导航、安全信息、机器人、医院、安全、工业、博物馆、物流、停车导航、建筑物测量和监控。 [Pos13 p1-p4]

Requirements:

中文：要求：

- accuracy: from mm to decameter;
- availability;
- coverage area;
- scalability;
 - 中文：可扩展性；
- output positions: 3D, 2D floor location, symbolic proximity;
 - 中文：输出位置：3D、2D 楼层位置、符号接近度；

- update rate;
- latency;
- number of users;
- cost. [Pos13 p5-p10]

User side / server side processing

中文：用户端/服务器端处理

①Centralized/server-side positioning: mobile positions are computed at a central facility/server.

Advantage: simplicity, output consistency and homogeneous processing. [Pos13 p18]

中文：集中/服务器端定位：移动位置在中央设施/服务器处计算。优点：简单、输出一致性和同质处理。 [Pos13 p18]

②User-side/distributed positioning: computation is performed on the user device. It is well suited when known nodes transmit and the mobile receives. [Pos14 p7]

中文：用户端/分布式定位：计算在用户设备上进行。它非常适合已知节点发送且移动设备接收的情况。 [Pos14 p7]

Known nodes, active and passive sensors

中文：已知节点、主动和被动传感器

Known nodes: fixed points whose coordinates are known, where markers or sensors for positioning unknown nodes are installed. [Pos13 p17]

中文：已知节点：坐标已知的固定点，安装有用于定位未知节点的标记或传感器。 [Pos13 p17]

Unknown nodes/mobiles: mobile users with sensors whose coordinates have to be estimated. [Pos13 p17]

中文：未知节点/移动设备：具有必须估计其坐标的传感器的移动用户。 [Pos13 p17]

Active/transmitter sensor: transmits signals.

Passive/receiver sensor: receives or reads a signal. [Pos13 p18]

有源/发射传感器：传输信号。

无源/接收传感器：接收或读取信号。 [Pos13 p18]

RSSI and ToA

RSSI: Received Signal Strength Indicator, it is used to receive signal strength averaged over a period, usually as power in db. [Pos14 p2]

中文：RSSI：接收信号强度指示器，一段时间内平均接收信号强度，通常以分贝为单位的功率。 [Pos14 p2]

Db formula:

中文：分贝公式：

$$dB = 10 \log_{10} \left(\frac{P}{P_0} \right)$$

Parameters:

- P : measured power.
 - 中文： P ：测量功率。
- P_0 : reference power. [Pos14 p2]
 - 中文： P_0 ：参考功率。 [Pos14 p2]

Physical attenuation model:

中文：物理衰减模型：THE physical attenuation is the basic formula, used to measure the distance of RSSI for inertial navigation 该物理衰减模型是室内定位中 RSSI 测距的基础公式，描述无线信号强度会随传播距离增加发生衰减。

$$P_R = k P_T \frac{G_T G_R}{4\pi d_{TR}^p}$$

Parameters:

中文：参数：

- P_R, P_T : received and transmitted signal strengths.
 - 中文： P_R, P_T ：接收和发送的信号强度。
- G_T, G_R : transmitter and receiver antenna gains.
 - 中文： G_T, G_R ：发射机和接收机天线增益。
- d_{TR} : distance between transmitter and receiver.
 - 中文： d_{TR} ：发射器和接收机之间的距离。
- p : path loss exponent.
 - 中文： p ：路径损耗指数。
- $p = 2$ in open space; indoor typically $4 \leq p \leq 6$, but in narrow corridors $p < 2$ because multipath cumulates. [Pos14 p3]
 - 中文： $p = 2$ 在开放空间；室内通常为 $4 \leq p \leq 6$ ，但在狭窄的走廊 $p < 2$ ，因为多径累积。 [Pos14 p3]

ToA: Time of Arrival, based on travel time from transmitter to receiver, like GNSS pseudorange.

Main technical issue: clock synchronization. [Pos14 p4]

中文：ToA：到达时间，基于从发射器到接收机的行程时间，如 GNSS 伪距。主要技术问题：时钟同步。 [Pos14 p4]

ToA equation:

$$ToA_i^u = \tau_i^u + \delta_u - \delta_i$$

Parameters:

- ToA_i^u : measured time of arrival between transmitter i and user u .
• 中文: ToA_i^u : 发射机 i 和用户 u 之间测量的到达时间。
- τ_i^u : signal travel time.
• 中文: τ_i^u : 信号传播时间。
- δ_u : receiver/user clock offset.
• 中文: δ_u : 接收机/用户钟差。
- δ_i : transmitter clock offset.
• 中文: δ_i : 发送器钟差。

If transmitters are known synchronized nodes:

中文: 如果发射机是已知的同步节点:

$$ToA_i^u = \tau_i^u + \delta_u$$

Then two or three coordinates plus one user clock offset must be solved. [Pos14 p4-p5]

中文: 然后必须求解两个或三个坐标加上一个用户钟差。 [Pos14 p4-p5]

Cell of Origin and Centroid原始小区与质心

Cell of Origin: the mobile is located at the position of the most probable known node, for example highest RSSI or smallest ToA. Accuracy depends on density of known points. [Pos14 p11]

中文: 原始小区: 移动设备位于最可能的已知节点的位置, 例如最高的 RSSI 或最小的 ToA。准确性取决于已知点的密度。 [Pos14 p11]

Centroid: the mobile is located in the centroid or weighted mean of sensed known nodes:

中文: 质心: 移动设备位于感测到的已知节点的质心或加权平均值:

$$\mathbf{x}_u = \sum_{i=1}^N w_i \mathbf{x}_i, \quad \sum_{i=1}^N w_i = 1$$

For RSSI, weights are proportional to observations. For ToA, weights are inversely proportional to observations. [Pos14 p12]

中文: 对于 RSSI, 权重与观测值成正比。对于 ToA, 权重与观测值成反比。 [Pos14 p12]

1.6 LBS

General description of LBS

LBS are **fully location-aware applications** delivering geolocated services through client-server or peer-to-peer architectures by wireless, cellular and satellite communication networks.

[Pos15 p3]

中文：LBS 是完全位置感知的应用程序，通过无线、蜂窝和卫星通信网络通过客户端-服务器或点对点架构提供地理定位/情境化服务。 [Pos15 p3]

it provides geographically oriented data and information services to users across mobile telecommunication networks. [Pos15 p3]

中文：其他PPT定义：通过移动通信网络向用户提供面向地理的数据和信息服务。 [Pos15 p3]

Users needs

Users are central to Location-Based Services, which must be designed from a user-centered perspective based on their needs and behaviors profiles.

Users determine which information should be provided and influence how systems are designed. User modeling asks:

1. who are the users?
2. what are their needs?
3. when and where do they need? [Pos15 p8]

Location and data modelling

Location modeling has two models:

Geographic referenced model地理参考模型：

- In the geographic model, locations and objects are represented as points, lines, areas and volumes consistently georeferenced in a reference system, with attributes.
- 中文：位置和对象是参考系统中一致地理参考的点、线、区域和体积，具有属性。
- Used for positions and proximity.
- 中文：用于位置和接近度。

Symbolic/conceptual model符号/概念模型：

In contrast, the symbolic model represents locations as classes with attributes and relationships: located objects belong to categories such as roads, boundaries, shops, hospitals, or hotels.

Together, these two models allow LBS to handle both geometric structure and semantic meaning语义含义。 .

Path optimization: requirements and costs

Path optimization is the process of finding the best feasible path between two locations in a network by minimizing a chosen cost, such as distance, time, or expense, under given topological constraints.

Roads are sequences of connections between intersections.

Intersections and terminal points are nodes. [Pos15 p14]

中文：道路是交叉路口之间的一系列连接。交点和终点是节点。 [Pos15 p14]

the Typical Costs can be:

- length;
- fee;
- time, possibly dynamic/on the fly. [Pos15 p14]

Topological rules:

- A road is split when different geometrical/topological conditions exist, such as speed, way rights, or costs.
 - 中文：当存在不同的几何/拓扑条件（例如速度、通行权或成本）时，道路就会被分割。
- Bridges/tunnels can geographically overlap but have no topological intersection, so no node is created between them.
 - 中文：桥梁/隧道可以在地理上重叠，但没有拓扑交叉点，因此它们之间不会创建任何节点。
- For each node, entering connections and allowed turns are stored, including turn costs. [Pos15 p15-p18]
 - 中文：对于每个节点，存储输入的连接和允许的转弯，包括转弯成本。 [Pos15 p15-p18]

2. Second level: improve the mark

2.1 RF

Helmert transformation in 3D

中文：3D 中的赫尔默特变换

General 3D Helmert form:

中文：一般 3D Helmert 形式：

$$\mathbf{x}_B = \mathbf{t} + (1 + s)R\mathbf{x}_A$$

Parameters:

中文：参数：

- $\mathbf{x}_A, \mathbf{x}_B$: coordinates in source and target frames.
 - 中文： $\mathbf{x}_A, \mathbf{x}_B$ ：源帧和目标帧中的坐标。
- $\mathbf{t} = [t_x, t_y, t_z]^T$: translations.
- s : scale factor.
- R : 3D rotation matrix, often approximated by three small rotations.
 - 中文： R ：3D 旋转矩阵，通常由三个小旋转来近似。

Precession, nutation and Chandler wobble

中文：进动、章动和钱德勒摆动

Precession: long-period motion of Earth's rotation axis. [Pos03 p3-p4]

进动：地球自转轴的长周期运动。 [Pos03 p3-p4]

Nutation: fluctuation around the precession motion, with angle about 9.2". [Pos03 p3-p4]

章动：围绕进动运动的波动，角度约为 9.2"。 [Pos03 p3-p4]

Because of precession and nutation, equator and ecliptic move in time. [Pos03 p6]

由于进动和章动，赤道和黄道会及时移动。 [Pos03 p6]

Chandler wobble / polar motion: the position of the pole moves with a period of about 435 days. [Pos03 p8]

钱德勒摆动/极移：极点的位置以大约 435 天为周期移动。 [Pos03 p8]

Possible expansion:

可能的扩展：

- These effects are part of Earth Orientation Parameters and are needed for transformations between celestial and terrestrial systems.
 - 中文：这些效应是地球定向参数的一部分，是天体和陆地系统之间转换所必需的。

Coordinates modeling in time for permanent stations of ITRF and ETRF 国际地球参考框架与欧洲地球参考框架永久观测站的时序坐标建模

Permanent station coordinates are not just fixed numbers; they are modeled in time using position at a reference epoch and velocity. [Pos03 p36-p38]

中文：永久站点坐标不仅仅是固定的数字；它们是使用参考历元的位置和速度对时间进行建模的。 [Pos03 p36-p38]

Basic model:

$$\mathbf{x}(t) = \mathbf{x}(t_0) + \dot{\mathbf{x}}(t - t_0)$$

Parameters:

- $\mathbf{x}(t)$: station coordinates at epoch t .
 - 中文： $\mathbf{x}(t)$ ： t 历元的台站坐标。
- $\mathbf{x}(t_0)$: coordinates at reference epoch.
 - 中文： $\mathbf{x}(t_0)$ ：参考历元的坐标。
- $\dot{\mathbf{x}}$: station velocity.
 - 中文： $\dot{\mathbf{x}}$ ：站速度。

- $t - t_0$: time interval.
- 中文: $t - t_0$: 时间间隔。

Local Level: differences between LC and LL

Local Cartesian (LC):

- East, North, Up axes.
 - 中文: 东、北、上轴。
- Up is ellipsoidal normal.
 - 中文: 向上为椭球法线。
- Used for navigation/local applications. [Pos03 p58-p60]
 - 中文: 用于导航/本地应用程序。 [Pos03 p58-p60]

Local Level (LL):

- Origin in point P_0 .
- Z axis oriented along gravity vertical.
 - 中文: Z 轴沿重力垂直方向定向。
- $X - Y$ plane is horizontal.
 - 中文: $X - Y$ 平面是水平的。
- It needs vertical deflection components (ξ, η) and a horizontal angle α to transform to LC. [Pos03 p69-p70]
 - 中文: 它需要垂直偏转分量 (ξ, η) 和水平角 α 才能转换为LC。 [Pos03 p69-p70]

Key difference: **LC is ellipsoid-based; LL is gravity-vertical based.**

中文: 主要区别: LC 是基于椭球体的; LL 基于重力垂直。

Diagram flow between local systems and geocentric systems

中文: 本地系统和地心系统之间的流程图

From PPT flow:

中文: 从PPT流程来看:

$$BF/LL \rightarrow LC \rightarrow GC \rightarrow (\varphi, \lambda, h)$$

Inputs:

- roll, pitch, yaw for BF to LC;
 - BF 到 LC 的横滚、俯仰、偏航;
- vertical deflection (ξ, η) and horizontal angle for LL to LC;
 - LL 至 LC 的垂直偏转 (ξ, η) 和水平角;

- origin of LC in GC;
 - GC中LC的起源;
- ellipsoid parameters (a, e) . [Pos03 p72]
 - 椭球参数 (a, e) 。 [Pos03 p72]

Covariance propagation in RF transformations

中文：RF 变换中的协方差传播

For a rotation transformation:

中文：对于旋转变换：

$$C_{new} = BC_{old}B^T$$

Parameters:

- C_{old} : covariance matrix in the original frame.
 - 中文： C_{old} ：原始帧中的协方差矩阵。
- B : rotation matrix from old frame to new frame.
 - 中文： B ：从旧帧到新帧的旋转矩阵。
- C_{new} : propagated covariance matrix.
 - 中文： C_{new} ：传播协方差矩阵。

2.2 GNSS

Keplerian orbits

A Keplerian orbit is stationary and defined by 6 parameters. [Pos06 p12]

中文：开普勒轨道是静止的，由 6 个参数定义。 [Pos06 p12]

The usual six parameters:

中文：常用的六个参数：

- a : semi-major axis.
 - 中文： a ：半长轴。
- e : eccentricity.
 - 中文： e ：偏心率。
- i : inclination.
 - 中文： i ：倾斜度。
- Ω : right ascension of ascending node.
 - 中文： Ω ：升交点赤经。
- ω : argument of perigee.

- 中文： ω ：近地点角。
- M_0 or anomaly at reference epoch.
 - 中文： M_0 或参考时期的异常。

Perturbation of actual orbits with respect to Keplerian

中文：实际轨道相对于开普勒的扰动

Actual GNSS orbits are not perfectly Keplerian because of perturbations. The control segment monitors satellites and predicts ephemerides. [Pos06 p22–p27]

中文：由于扰动，实际的 GNSS 轨道并不完全是开普勒轨道。控制部分监视卫星并预测星历表。 [Pos06 p22–p27]

Perturbations require corrections and periodic updates of broadcast ephemerides.

中文：扰动需要广播星历表的修正和定期更新。

Broadcast ephemerides

中文：广播星历表

Satellite ephemerides are parameters that allow computation of satellite position. Broadcast ephemerides are transmitted in the navigation message and allow real-time positioning. [Pos06 p4, p27]

中文：卫星星历表是允许计算卫星位置的参数。广播星历表在导航消息中传输并允许实时定位。 [Pos06 p4, p27]

PPT values:

中文：PPT值：

- Broadcast ephemerides accuracy is about 1 m.
 - 中文：广播星历精度约为1m。
- GPS broadcast quasi-Keplerian parameters are updated every 2 hours. [Pos06 p27, p29]
 - 中文：GPS广播准开普勒参数每2小时更新一次。 [Pos06 p27, p29]

Oscillator/美 /'ɑ:sɪleɪtər/: frequency, phase and time

中文：振荡器：频率、相位和时间

Carrier and code generation is based on oscillator signals. For a sinusoidal oscillator:

中文：载波和代码生成基于振荡器信号。对于正弦振荡器：

$$\Phi(t) = ft + \int f dt(t) + \Phi_0$$

Parameters:

- $\Phi(t)$: phase at time t .
- f : oscillator frequency.
- $dt(t)$: clock offset in time.
- Φ_0 : initial phase. [Pos10 p6]
 - 中文: Φ_0 : 初始阶段。 [Pos10 p6]

Observation equation of codes / pseudoranges

中文: 代码/伪距观测方程

Use the SPP code equation from section 1.3. It includes geometric range, receiver/satellite clocks, hardware biases, troposphere, ionosphere, and noise. [Pos09 p2]

中文: 使用 1.3 节中的 SPP 代码方程。它包括几何范围、接收机/卫星时钟、硬件偏差、对流层、电离层和噪声。 [Pos09 p2]

For linearized SPP:

中文: 对于线性化 SPP:

$$\Delta P_R^S = \mathbf{e}_R^S \Delta \mathbf{x}_R + c dt_R + \epsilon_R^S + \nu_R^S$$

Parameters:

中文: 参数:

- ΔP_R^S : reduced code observation, observation minus known terms.
 - 中文: ΔP_R^S : 简化后的码观测, 观测值减去已知项。
- $\mathbf{e}_R^S$: line-of-sight unit vector from receiver to satellite.
 - 中文: $\mathbf{e}_R^S$: 从接收机到卫星的视距单位矢量。
- $\Delta \mathbf{x}_R = [\Delta X_R, \Delta Y_R, \Delta Z_R]^T$: correction to approximate receiver coordinates.
 - 中文: $\Delta \mathbf{x}_R = [\Delta X_R, \Delta Y_R, \Delta Z_R]^T$: 修正近似接收机坐标。
- $c dt_R$: receiver clock unknown in meters.
 - 中文: $c dt_R$: 接收机时钟未知 (米)。
- ϵ_R^S : residual bias from atmosphere, satellite coordinates and clock modeling.
 - 中文: ϵ_R^S : 来自大气、卫星坐标和时钟模型的残余偏差。
- ν_R^S : observation noise. [Pos09 p11-p13]
 - 中文: ν_R^S : 观察噪声。 [Pos09 p11-p13]

Observation equation of phases

中文: 相观测方程

Carrier phase observation:

中文: 载波相位观测:

$$\Phi_{R,i}^{S,G}(t) = \rho_R^S(t) + c[dt_R(t) + b_{R,i}^G - dt^S(t) - b_i^{S,G}] + T_R^S(t) - I_{R,i}^S(t) + \lambda_i N_{R,i}^{S,G} + \varepsilon_\Phi$$

Parameters:

中文：参数：

- $\Phi_{R,i}^{S,G}$: carrier phase observation in meters.
 - 中文： $\Phi_{R,i}^{S,G}$ ：以米为单位的载波相位观测。
- ρ_R^S : geometric range.
 - 中文： ρ_R^S ：几何范围。
- c : speed of light.
 - 中文： c ：光速。
- dt_R, dt^S : receiver and satellite clock offsets.
 - 中文： dt_R, dt^S ：接收机和卫星钟差。
- $b_{R,i}^G, b_i^{S,G}$: phase hardware biases.
 - 中文： $b_{R,i}^G, b_i^{S,G}$ ：相位硬件偏差。
- T_R^S : tropospheric delay.
 - 中文： T_R^S ：对流层延迟。
- $I_{R,i}^S$: ionospheric delay, opposite sign with respect to codes.
 - 中文： $I_{R,i}^S$ ：电离层延迟，与代码符号相反。
- λ_i : carrier wavelength.
 - 中文： λ_i ：载波波长。
- $N_{R,i}^{S,G}$: integer ambiguity.
 - 中文： $N_{R,i}^{S,G}$ ：整数歧义。
- ε_Φ : phase noise, about 1-2 mm. [Pos10 p5-p9]
 - 中文： ε_Φ ：相位噪声，约1-2毫米。 [Pos10 p5-p9]

Initial ambiguities

The receiver can observe the fractional part of the incoming carrier, but not the integer number of cycles from satellite emission to receiver epoch. This unknown integer is the ambiguity.

[Pos10 p4]

中文：接收机可以观察到输入载波的小数部分，但不能观察到从卫星发射到接收机历元的整数周期数。这个未知的整数就是歧义性。 [Pos10 p4]

If no loss of lock happens, the initial ambiguity is unknown but constant in time. With enough epochs, redundancy is reached and a system can solve coordinates and ambiguities. [Pos10 p13]

中文：如果没有发生锁丢失，则初始模糊度是未知的，但随着时间的推移是恒定的。有了足够的纪元，就达到了冗余，系统可以解决坐标和模糊度。 [Pos10 p13]

If signal lock is lost, a new ambiguity appears; this is equivalent to a cycle slip. [Pos10 p30]

中文：如果信号锁定丢失，就会出现新的歧义；这相当于周跳。 [Pos10 p30]

Ionospheric delay in detail

中文：电离层延迟详细信息

The ionosphere is due to free electrons between 100 and 1000 km altitude. Its refraction index is frequency dependent. [Pos08 p5]

中文：电离层是由 100 至 1000 公里高程之间的自由电子形成的。其折射率与频率相关。

[Pos08 p5]

Simplified PPT formula:

$$n(x) = 1 \pm \frac{A N_e(x)}{f^2}$$
$$I_R^S = \pm \frac{A}{f^2} \int_R^S N_e(x) dx = \pm \frac{A}{f^2} TEC_R^S$$

Parameters:

- $n(x)$: refraction index along the signal path.
 - 中文： $n(x)$ ：沿信号路径的折射率。
- A : physical constant.
 - 中文： A ：物理常数。
- $N_e(x)$: local electron density, in electrons/m³.
 - 中文： $N_e(x)$ ：局域电子密度，单位为电子/m³。
- f : carrier frequency in Hz.
 - 中文： f ：载波频率（以 Hz 为单位）。
- TEC : Total Electron Content, free electron surface density along the ionospheric path.
 - 中文： TEC ：总电子含量，沿电离层路径的自由电子表面密度。
- Sign: positive for codes/group delay, negative for carrier phase. [Pos08 p5-p7]
 - 中文：符号：正值表示代码/群延迟，负值表示载波相位。 [Pos08 p5-p7]

PPT values:

- At 15 degrees elevation, ionospheric delay can range from 3.9 m at night to 180 m under intense solar activity.
 - 中文：在海拔 15 度时，电离层延迟范围可以从夜间的 3.9 m 到强烈太阳活动时的 180 m。
- 1 TECU is the TEC measurement unit. [Pos08 p8]
 - 中文：1 TECU 是 TEC 测量单位。 [Pos08 p8]

Models transmitted:

中文：传输型号：

- GPS: Klobuchar.
- GLONASS: no ionospheric model.
- BeiDou: BDGIM.
- Galileo: NeQuick. [Pos08 p9]

Ionospheric-free combination for codes and phases

中文：代码和相位的无电离层组合

For two frequencies f_1, f_2 , define:

中文：对于两个频率 f_1, f_2 ，定义：

$$\alpha_1 = \frac{f_1^2}{f_1^2 - f_2^2}, \quad \alpha_2 = -\frac{f_2^2}{f_1^2 - f_2^2}$$

Code ionosphere-free combination:

中文：代码无电离层组合：

$$P_3 = \alpha_1 P_1 + \alpha_2 P_2$$

Phase ionosphere-free combination:

中文：无相位电离层组合：

$$L_3 = \alpha_1 L_1 + \alpha_2 L_2$$

Parameters:

- P_1, P_2 : code observations on frequencies f_1, f_2 .
 - 中文： P_1, P_2 ：在频率 f_1, f_2 的码观测。
- L_1, L_2 : phase observations on frequencies f_1, f_2 .
 - 中文： L_1, L_2 ：频率 f_1, f_2 的相位观测。
- P_3, L_3 : ionosphere-free combined observations.
 - 中文： P_3, L_3 ：无电离层组合观测。
- α_1, α_2 : coefficients chosen to cancel first-order ionospheric delay.
 - 中文： α_1, α_2 ：选择用于消除一阶电离层延迟的系数。

PPT result: no ionospheric delay on the combined observation. [Pos09 p3-p5, Pos10 p14]

中文：PPT结果：联合观测无电离层延迟。 [Pos09 p3-p5, Pos10 p14]

Tropospheric delay in detail

中文：对流层延迟详细信息

Tropospheric effect is due to air and water vapor content from 0 to 40 km height. [Pos08 p14]

中文：对流层效应是由于0至40公里高程的空气和水蒸气含量造成的。 [Pos08 p14]

It has:

中文：它有：

- dry component: about 90 percent of the total delay;
 - 中文：干成分：约占总延迟的90%；
- wet component. [Pos08 p15]
 - 中文：湿成分。 [Pos08 p15]

Standard models exist, for example Saastamoinen model, based on ideal hypotheses on the atmosphere. [Pos08 p16]

中文：存在基于对大气的理想假设的标准模型，例如 Saastamoinen 模型。 [Pos08 p16]

Zenith Tropospheric Delay (ZTD): tropospheric delay at the zenith of a receiver. [Pos08 p16]

中文：天顶对流层延迟 (ZTD)：接收机天顶处的对流层延迟。 [Pos08 p16]

PPT key warning: tropospheric models never completely represent the real meteorological state during the survey. Any modeling of T leaves residual error, about 5 percent of total delay.

[Pos08 p19]

中文：PPT重点警告：对流层模型永远不能完全代表测量过程中的真实气象状况。 T 的任何建模都会留下残余误差，约占总延迟的 5%。 [Pos08 p19]

PPP residual troposphere model:

中文：PPP残余对流层模型：

$$T_R^S(t) = T_{0,R}^S(t) + \delta T_R^S(t)$$

$$\delta T_R^S(t) = \delta T_Z(t) m(el_R^S)$$

Parameters:

中文：参数：

- $T_R^S(t)$: total tropospheric delay.
 - 中文： $T_R^S(t)$ ：对流层总延迟。
- $T_{0,R}^S(t)$: standard model value.
 - 中文： $T_{0,R}^S(t)$ ：标准型号值。
- $\delta T_R^S(t)$: residual/model error along satellite direction.
 - 中文： $\delta T_R^S(t)$ ：沿卫星方向的残差/模型误差。
- $\delta T_Z(t)$: zenith tropospheric residual, estimated parameter.
 - 中文： $\delta T_Z(t)$ ：天顶对流层残差，估计参数。
- $m(el_R^S)$: mapping function depending on satellite elevation.

- 中文: $m(el_R^S)$: 取决于卫星高程的映射函数。
- el_R^S : satellite elevation. [Pos10 p16–p17]
 - 中文: el_R^S : 卫星高程。 [Pos10 p16–p17]

Diagram flow of PPP

中文: PPP流程示意图

Static PPP scheme:

中文: 静态PPP方案:

1. multi-frequency phase observations;
中文: 1.多频相位观测;
2. precise ephemerides, clock offsets, hardware biases;
中文: 2. 精确星历、钟差、硬件偏差;
3. ionosphere-free combination;
中文: 3.无电离层组合;
4. tropospheric residual estimation;
中文: 4.对流层残差估计;
5. batch processing over many epochs;
中文: 5. 多个epoch的批处理;
6. estimate coordinates, receiver clocks/hardware biases, ambiguities, tropospheric parameters. [Pos10 p15–p20, p23–p24]
中文: 6. 估计坐标、接收机时钟/硬件偏差、模糊度、对流层参数。 [Pos10 p15–p20, p23–p24]

Single and double differences

中文: 单差和双差

Relative positioning is mainly implemented on carrier phases for geodetic applications. [Pos11 p2]

中文: 相对定位主要在大地测量应用的载波相位上实现。 [Pos11 p2]

Single difference between two receivers to the same satellite reduces satellite-related errors and satellite clock effects.

中文: 同一卫星的两个接收机之间的单一差异减少了与卫星相关的误差和卫星时钟效应。

Double difference between two receivers and two satellites further removes receiver clock effects and reduces common errors. The remaining important errors are:

中文: 两个接收机和两颗卫星之间的双差进一步消除了接收机时钟效应并减少了常见错误。其余的重要错误是:

- differential ephemerides error;
 - 中文：差分星历误差；
- differential tropospheric effect;
 - 中文：对流层微分效应；
- differenced multipath;
 - 中文：差分多径；
- ambiguity terms. [Pos11 p3-p5, p14-p16]
 - 中文：歧义术语。 [Pos11 p3-p5, p14-p16]

DD biases and error budget in single epoch

中文：单个 epoch 中的 DD 偏差和误差预算

Remaining DD errors:

中文：剩余 DD 错误：

- ephemerides error;
 - 中文：星历表错误；
- tropospheric model error;
 - 中文：对流层模型误差；
- ionospheric model error on single frequency;
 - 中文：单频电离层模型误差；
- multipath. [Pos11 p16-p19]
 - 中文：多路径。 [Pos11 p16-p19]

For short baselines, common atmospheric and orbit errors are strongly reduced; for longer baselines, differential residuals increase.

中文：对于短基线，常见的大气和轨道误差大大减少；对于较长的基线，微分残差会增加。

Phase DD batch processing and accuracies

中文：相位 DD 批处理和精度

Phase DD batch processing uses all epochs and stations, with ambiguities and coordinates estimated. Complete ambiguity fixing is essential for the best accuracies. [Pos11 p34-p39]

中文：DD 阶段批处理使用所有历元和站点，并估计模糊度和坐标。完整的歧义修复对于获得最佳准确性至关重要。 [Pos11 p34-p39]

Expected final accuracies:

中文：预期最终精度：

- Kinematic phase DD: centimeter level when conditions are good.

- 中文：运动相位DD：条件好的情况下为厘米级。
- Static phase DD: millimeter to centimeter level for long sessions and fixed ambiguities. [Pos11 p35-p39]
 - 中文：静态阶段 DD：毫米到厘米级别，用于长时间会话和固定模糊度。 [Pos11 p35-p39]

2.3 INS

中文：2.3 惯性导航

Principle and equations of open loop spring accelerometers

中文：开环弹簧加速度计的原理和方程

Open loop spring accelerometers measure proof mass displacement due to external forces. [Pos12 p7-p8]

中文：开环弹簧加速度计测量由于外力引起的质量块位移。 [Pos12 p7-p8]

For constant force after initial oscillations:

中文：对于初始振荡后的恒定力：

$$F = ka$$

PPT conclusion:

$$a = \frac{F}{m}$$

Parameters:

中文：参数：

- F : force acting on proof mass along spring axis.
 - 中文： F ：沿弹簧轴作用在检验质量上的力。
- m : proof mass.
 - 中文： m ：证明质量。
- a : acceleration along spring axis.
 - 中文： a ：沿弹簧轴的加速度。

The full spring dynamics includes mass, damping, stiffness and external force; the teacher may focus on the physical principle and the final relation. [Pos12 p8]

中文：全弹簧动力学包括质量、阻尼、刚度和外力；老师可以重点讲物理原理和最终的关系。 [Pos12 p8]

Principle and equations of optical gyroscopes

中文：光学陀螺仪原理及方程

Fiber optic gyroscopes are based on Sagnac effect. Two light beams travel in opposite directions in a rotating fiber coil; the phase shift is proportional to angular rate. [Pos12 p13]

中文：光纤陀螺仪基于萨尼亚克效应。两束光束在旋转光纤线圈中沿相反方向传播；相移与角速率成正比。 [Pos12 p13]

PPT formula:

$$\Delta\Phi = \frac{4\Omega A}{\lambda c}$$

Parameters:

中文：参数：

- $\Delta\Phi$: phase shift between opposite light beams.
 - 中文： $\Delta\Phi$ ：相对光束之间的相移。
- Ω : angular rate.
 - 中文： Ω ：角速率。
- A : area enclosed by the optical path.
 - 中文： A ：光路包围的区域。
- λ : wavelength.
 - 中文： λ ：波长。
- c : speed of light.
 - 中文： c ：光速。

Typical error budget of IMU and INS

中文：IMU 和 INS 的典型误差预算

Accelerometer error model:

中文：加速度计误差模型：

$$\delta a = b + \lambda a + c_T \Delta T + v$$

Gyroscope error model:

中文：陀螺仪误差模型：

$$\delta\omega = b + \lambda\omega + c_T \Delta T + v$$

Parameters:

中文：参数：

- b : bias.
 - 中文： b ：偏差。

- λ : scale factor.
 - 中文: λ : 比例因子。
- c_T : thermal constant depending on temperature.
 - 中文: c_T : 取决于温度的热常数。
- ΔT : temperature variation.
 - 中文: ΔT : 温度变化。
- v : measurement noise. [Pos12 p11, p14]
 - 中文: v : 测量噪声。 [Pos12 p11, p14]

Rotations and differential equations

中文: 旋转和微分方程

In strapdown systems, accelerometers measure specific acceleration in the body frame:

中文: 在捷联系统中, 加速度计测量车身框架中的特定加速度:

$$\mathbf{f}^i = R_b^i \mathbf{f}^b$$

Parameters:

中文: 参数:

- $\mathbf{f}^b$: specific acceleration measured in body frame.
 - 中文: $\mathbf{f}^b$: 在车身框架中测量的特定加速度。
- $\mathbf{f}^i$: specific acceleration in inertial frame.
 - 中文: $\mathbf{f}^i$: 惯性系中的特定加速度。
- R_b^i : rotation matrix from body to inertial frame.
 - 中文: R_b^i : 从物体到惯性系的旋转矩阵。

The rotation matrix is derived by integrating gyroscope measurements:

中文: 旋转矩阵是通过积分陀螺仪测量得出的:

$$\dot{R}_b^i = R_b^i \Omega_{ib}^b$$

where Ω_{ib}^b is the skew-symmetric matrix of angular rate measured by gyroscopes. [Pos12 p20]

中文: 其中 Ω_{ib}^b 是陀螺仪测量的角速率的斜对称矩阵。 [Pos12 p20]

Solution of differential equation in 2D

中文: 二维微分方程的解

For a single-axis rotation in 2D, the rotation matrix is:

中文: 对于二维单轴旋转, 旋转矩阵为:

$$R = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The differential equation requires initial attitude. [Pos12 p21]

中文：微分方程需要初始姿态。 [Pos12 p21]

2.4 Indoor positioning second level

中文：2.4 室内定位二级

Observation equations of RSSI and ToA

中文：RSSI和ToA观测方程

PPT generic parametric observation equations:

中文：PPT通用参数观测方程：

$$\frac{P_T}{P_R} = K d_{TR}^p$$

$$\Delta t_{TR} = \frac{d_{TR} + l}{c}$$

Parameters:

中文：参数：

- P_T, P_R : transmitted and received power.
 - 中文： P_T, P_R ：发射和接收功率。
- K : proportionality constant.
 - 中文： K ：比例常数。
- d_{TR} : distance between transmitter and receiver.
 - 中文： d_{TR} ：发射器和接收机之间的距离。
- p : path loss exponent depending on local conditions.
 - 中文： p ：路径损耗指数取决于当地条件。
- Δt_{TR} : travel time observation.
 - 中文： Δt_{TR} ：行程时间观察。
- l : excess path, depending on local conditions.
 - 中文： l ：多余路径，视当地情况而定。
- c : signal propagation speed. [Pos14 p9]
 - 中文： c ：信号传播速度。 [Pos14 p9]

Fingerprinting

中文：指纹识别

Fingerprinting has two steps:

中文：指纹识别有两个步骤：

1. calibration;

中文：1.校准；

2. usage. [Pos14 p14]

中文：2.用途。 [Pos14 p14]

Calibration:

中文：校准：

- choose M control points in the area;
 - 中文：选择区域内的 M 控制点；
- estimate their coordinates accurately;
 - 中文：准确估计它们的坐标；
- in each control point store signals from/to all N known nodes.
 - 中文：在每个控制点中存储来自/到所有 N 已知节点的信号。

Stored data for each control point:

中文：每个控制点存储的数据：

$$\mathbf{x}_{CP_i} = [x_{CP_i}, y_{CP_i}]^T, \quad \mathbf{r}_{CP_i} = [r_{KN1}, r_{KN2}, \dots, r_{KNN}]^T$$

Usage:

中文：用法：

- at unknown position, the mobile measures a signal vector $\mathbf{s}(t_u)$;
 - 中文：在未知位置，移动设备测量信号矢量 $\mathbf{s}(t_u)$ ；
- compare it with all stored fingerprints;
 - 中文：与所有存储的指纹进行比较；
- assign the position of the control point minimizing the difference:
 - 中文：分配控制点的位置以最小化差异：

$$\mathbf{x}_u = \mathbf{x}_{CP_i} \quad \text{such that} \quad \|\mathbf{s}(t_u) - \mathbf{r}_{CP_i}\| = \min$$

Problems:

中文：问题：

- users may have different devices from calibration devices;
 - 中文：用户可能拥有与校准设备不同的设备；
- furniture replacement and people traffic change signal geometry. [Pos14 p14-p16]

- 中文：家具更换和人流改变信号几何形状。 [Pos14 p14-p16]

WLAN/Wi-Fi and RFID

中文：WLAN/Wi-Fi 和 RFID

Wi-Fi:

中文：无线上网：

- IEEE 802.11 wireless local area networks.
 - 中文：IEEE 802.11 无线局域网。
- Frequencies 2-5 GHz.
 - 中文：频率 2-5 GHz。
- Range up to 50-100 m.
 - 中文：范围可达 50-100 m。
- Most popular positioning: RSSI; less common: ToA, DToA.
 - 中文：最流行的定位：RSSI；不太常见：ToA、DToA。
- CoO accuracy about 10-50 m; fingerprinting is promising but more expensive. [Pos14 p17]
 - 中文：CoO精度约10-50 m；指纹识别很有前途，但价格更高。 [Pos14 p17]

RFID:

中文：射频识别：

- Cheap technique to identify objects carrying RFID sensors.
 - 中文：识别携带 RFID 传感器的物体的廉价技术。
- Readers read signals from tags.
 - 中文：阅读器从标签读取信号。
- Tags can be passive or active.
 - 中文：标签可以是被动的或主动的。
- Passive tags backscatter reader signal, practically eternal, range limited to about 2 m.
 - 中文：无源标签反向散射阅读器信号，几乎是永恒的，范围限制在 2 m 左右。
- Active transceivers need power but have larger range. [Pos14 p18-p19]
 - 中文：有源收发器需要电源，但范围更大。 [Pos14 p18-p19]

2.5 LBS second level

中文：2.5 LBS二级

Database of routes and connections: topological rules

中文：路线和连接数据库：拓扑规则

Data stored for connections:

中文：为连接存储的数据：

- ID, name, starting node, ending node;
 - 中文：ID、名称、起始节点、结束节点；
- allowed direction S to E / E to S;
 - 中文：允许方向 S 至 E / E 至 S；
- costs such as time, length, fee. [Pos15 p16]
 - 中文：成本，例如时间、长度、费用。 [Pos15 p16]

Data stored for nodes:

中文：节点存储的数据：

- node ID;
 - 中文：节点ID；
- coordinates X, Y ;
 - 中文：坐标 X, Y ；
- number of connected connections. [Pos15 p17]
 - 中文：连接的连接数。 [Pos15 p17]

Data stored for turns:

中文：轮次存储的数据：

- node ID;
 - 中文：节点ID；
- pair of connected roads;
 - 中文：一对相连的道路；
- whether turn is allowed;
 - 中文：是否允许转弯；
- turn costs. [Pos15 p18]
 - 中文：转向成本。 [Pos15 p18]

3. Third level: full mark

3.1 Kalman filtering

中文：3.1 Kalman滤波

Purpose:

中文：目的：

1. predict the evolution of a dynamic system in real time;
中文：1.实时预测动态系统的演化；
2. update the previous estimate;
中文：2.更新之前的估算；
3. exploit new observations and the dynamics of the system. [PosApp2 p2]
中文：3. 利用新的观察结果和系统的动态。 [PosApp2 p2]

Dynamic model:

中文：动态模型：

$$\mathbf{x}_{t+1} = T_{t+1}\mathbf{x}_t + \xi_{t+1}$$

Observation model:

中文：观察模型：

$$\mathbf{y}_{t+1} = A_{t+1}\mathbf{x}_{t+1} + \nu_{t+1}$$

Parameters:

中文：参数：

- $\mathbf{x}_t$: state vector, e.g. position, velocity, acceleration, attitude.
 - 中文： $\mathbf{x}_t$ ：状态向量，例如位置、速度、加速度、姿态。
- T_{t+1} : transition matrix.
 - 中文： T_{t+1} ：转移矩阵。
- ξ_{t+1} : model error.
 - 中文： ξ_{t+1} ：模型错误。
- $\mathbf{y}_{t+1}$: observation vector.
 - 中文： $\mathbf{y}_{t+1}$ ：观察向量。
- A_{t+1} : design/observation matrix.
 - 中文： A_{t+1} ：设计/观察矩阵。
- ν_{t+1} : observation noise. [PosApp2 p3-p4]
 - 中文： ν_{t+1} ：观察噪声。 [PosApp2 p3-p4]

Prediction:

$$\mathbf{x}_{t+1|t} = T_{t+1}\mathbf{x}_{t|t}$$

Innovation:

$$\eta_{t+1} = \mathbf{y}_{t+1} - A_{t+1}\mathbf{x}_{t+1|t}$$

Update:

$$\mathbf{x}_{t+1|t+1} = \mathbf{x}_{t+1|t} + G_{t+1}\eta_{t+1}$$

Gain:

$$G_{t+1} = K_{t+1} A_{t+1}^T (C_\nu + A_{t+1} K_{t+1} A_{t+1}^T)^{-1}$$

Covariance prediction:

$$K_{t+1} = C_{\xi,t+1} + T_{t+1} C_{e,t} T_{t+1}^T$$

Covariance update:

$$C_{e,t+1} = K_{t+1} - G_{t+1} A_{t+1} K_{t+1}$$

Parameters:

- η_{t+1} : innovation, observation minus predicted observation.
 - 中文: η_{t+1} : 创新, 观察减去预测观察。
- G_{t+1} : Kalman gain.
 - 中文: G_{t+1} : Kalman增益。
- K_{t+1} : prediction error covariance.
 - 中文: K_{t+1} : 预测误差协方差。
- C_ν : observation noise covariance.
 - 中文: C_ν : 观测噪声协方差。
- $C_{\xi,t+1}$: model error covariance.
 - 中文: $C_{\xi,t+1}$: 模型误差协方差。
- $C_{e,t}$: estimation error covariance at epoch t . [PosApp2 p7-p13]
 - 中文: $C_{e,t}$: 第 t 时期的估计误差协方差。 [PosApp2 p7-p13]

3.2 RF: transformation between global Cartesian and geodetic coordinates

中文: 3.2 RF: 全局直角坐标坐标与大地坐标之间的转换

Forward transformation is in section 1.2.

中文: 正向变换在 1.2 节中。

Inverse transformation $[X, Y, Z] \rightarrow [\varphi, \lambda, h]$ requires computation but is well defined. [Pos03 p58]

中文: 逆变换 $[X, Y, Z] \rightarrow [\varphi, \lambda, h]$ 需要计算, 但定义明确。 [Pos03 p58]

Oral answer structure:

1. Compute longitude:
 - 中文: 1. 计算经度:

$$\lambda = \text{atan2}(Y, X)$$

2. Iteratively compute latitude φ because R_N depends on φ .

中文：2. 迭代计算纬度 φ ，因为 R_N 取决于 φ 。

3. Compute height:

中文：3. 计算高程：

$$h = \frac{\sqrt{X^2 + Y^2}}{\cos \varphi} - R_N$$

3.3 RF: transformations between Global Cartesian and Local Cartesian

中文：3.3 RF：全局直角坐标和局部直角坐标之间的转换

Given origin $P_0(\varphi_0, \lambda_0, h_0)$, compute geocentric baseline:

中文：给定原点 $P_0(\varphi_0, \lambda_0, h_0)$ ，计算地心基线：

$$\Delta \mathbf{x}_{GC} = \mathbf{x}_P - \mathbf{x}_{P_0}$$

Transform to local East-North-Up:

中文：转换为当地的东向北向上：

$$\mathbf{x}_{LC} = R_{GC}^{LC} \Delta \mathbf{x}_{GC}$$

where:

中文：在哪里：

$$R_{GC}^{LC} = \begin{bmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{bmatrix}$$

Parameters:

中文：参数：

- $\mathbf{x}_{LC} = [E, N, U]^T$.
 - 中文： $\mathbf{x}_{LC} = [E, N, U]^T$ 。
- $\Delta \mathbf{x}_{GC}$: geocentric baseline.
 - 中文： $\Delta \mathbf{x}_{GC}$ ：地心基线。
- φ_0, λ_0 : geodetic latitude/longitude of local origin.
 - 中文： φ_0, λ_0 ：当地原点的大地纬度/经度。
- R_{GC}^{LC} : rotation from global Cartesian to local Cartesian. [Pos03 p60–p62]
 - 中文： R_{GC}^{LC} ：从全局直角坐标坐标旋转到局部直角坐标坐标。 [Pos03 p60–p62]

3.4 GNSS: single epoch point positioning by Least Squares and final error budget

中文：3.4 GNSS：最小二乘单历元定位和最终误差预算

Observation model:

中文：观察模型：

$$\Delta \mathbf{P} = E \begin{bmatrix} \Delta X_R \\ \Delta Y_R \\ \Delta Z_R \\ c dt_R \end{bmatrix} + \boldsymbol{\epsilon} + \boldsymbol{\nu}$$

Where each row of E contains the line-of-sight components and 1 for the clock term:

中文：其中 E 的每一行包含视线分量，1 代表时钟项：

$$E_s = [e_x^s, e_y^s, e_z^s, 1]$$

Least Squares solution:

中文：最小二乘解：

$$\hat{\Delta \mathbf{x}} = (E^T Q^{-1} E)^{-1} E^T Q^{-1} \Delta \mathbf{P}$$

Parameters:

中文：参数：

- $\Delta \mathbf{P}$: vector of reduced code observations.
 - 中文： $\Delta \mathbf{P}$ ：简化码观测向量。
- E : design matrix.
 - 中文： E ：设计矩阵。
- Q : observation cofactor/weight matrix.
 - 中文： Q ：观察辅因子/权重矩阵。
- $\Delta X_R, \Delta Y_R, \Delta Z_R$: coordinate corrections.
 - 中文： $\Delta X_R, \Delta Y_R, \Delta Z_R$ ：坐标修正。
- $c dt_R$: receiver clock correction in meters.
 - 中文： $c dt_R$ ：接收机时钟校正（以米为单位）。
- $\boldsymbol{\epsilon}$: residual model biases.
 - 中文： $\boldsymbol{\epsilon}$ ：残差模型偏差。
- $\boldsymbol{\nu}$: observation noise. [Pos09 p9-p13, PosApp1 p13-p15]
 - 中文： $\boldsymbol{\nu}$ ：观察噪声。 [Pos09 p9-p13, PosApp1 p13-p15]

Final SPP error budget: see section 1.3.

中文：最终 SPP 误差预算：参见第 1.3 节。

3.5 GNSS: residual troposphere estimation in PPP

中文：3.5 GNSS：PPP 中的残余对流层估计

PPP estimates residual troposphere instead of only modeling it:

中文：PPP 估计残余对流层而不是仅对其进行建模：

$$T_R^S(t) = T_{0,R}^S(t) + \delta T_Z(t)m(el_R^S)$$

The zenith residual $\delta T_Z(t)$ is modeled in time as stepwise parameters and estimated together with coordinates, clock offsets, and ambiguities. [Pos10 p16-p17]

中文：天顶残差 $\delta T_Z(t)$ 在时间上被建模为逐步参数，并与坐标、钟差和模糊度一起估计。 [Pos10 p16-p17]

3.6 GNSS: ambiguity fixing

中文：3.6 GNSS：模糊度修正

Ambiguity fixing means estimating the integer phase ambiguities and fixing them to integer values. It is essential in high-precision carrier phase processing.

中文：模糊度固定意味着估计整数相位模糊度并将其固定为整数值。它对于高精度载波相位处理至关重要。

PPT points:

中文：PPT要点：

- In single epoch, each satellite introduces one ambiguity, so phases cannot be used alone in single epoch.
 - 中文：在单个历元中，每颗卫星都会引入一个模糊度，因此在单个历元中不能单独使用相位。
- If no loss of lock occurs, the initial ambiguity remains constant in time.
 - 中文：如果没有发生锁丢失，则初始模糊度随时间保持不变。
- With multiple epochs, redundancy allows solving coordinates and ambiguities.
 - 中文：对于多个纪元，冗余允许解决坐标和模糊度。
- Cycle slips introduce new ambiguities and must be detected. [Pos10 p11-p13, p30]
 - 中文：周跳带来了新的歧义，必须予以检测。 [Pos10 p11-p13, p30]

In phase DD processing, complete ambiguity fixing is essential to obtain best final accuracies. [Pos11 p39]

中文：在相位 DD 处理中，完整的模糊度修复对于获得最佳的最终精度至关重要。 [Pos11 p39]

3.7 INS: navigation equations from body system to Earth-fixed system

中文：3.7 INS：从本体系统到地固系统的导航方程

Navigation equations link unknowns, namely position, velocity and attitude, to accelerometer and gyroscope observations. [Pos12 p18]

中文：导航方程将未知数（即位置、速度和姿态）与加速度计和陀螺仪观测结果联系起来。
[Pos12 p18]

In Earth-fixed reference system, using Coriolis theorem:

中文：在地固参考系中，利用科里奥利定理：

$$\ddot{\mathbf{x}}^e = \mathbf{f}^e + \mathbf{g}^e - 2\Omega_{ie}^e \dot{\mathbf{x}}^e - \Omega_{ie}^e \Omega_{ie}^e \mathbf{x}^e$$

Strapdown body-to-Earth acceleration:

$$\mathbf{f}^e = R_i^e R_b^i \mathbf{f}^b$$

Navigation equations:

$$\frac{d\mathbf{x}^e}{dt} = \dot{\mathbf{x}}^e$$

$$\frac{d\dot{\mathbf{x}}^e}{dt} = R_i^e R_b^i \mathbf{f}^b + \mathbf{g}^e - 2\Omega_{ie}^e \dot{\mathbf{x}}^e$$

$$\dot{R}_b^i = R_b^i \Omega_{ib}^b$$

Parameters:

- $\mathbf{x}^e$: position in Earth-fixed reference frame.
 - 中文： $\mathbf{x}^e$ ：地固参考系中的位置。
- $\dot{\mathbf{x}}^e$: velocity in Earth-fixed frame.
 - 中文： $\dot{\mathbf{x}}^e$ ：地固坐标系中的速度。
- $\ddot{\mathbf{x}}^e$: acceleration in Earth-fixed frame.
 - 中文： $\ddot{\mathbf{x}}^e$ ：地固坐标系中的加速度。
- $\mathbf{f}^b$: specific acceleration measured by accelerometers in body frame.
 - 中文： $\mathbf{f}^b$ ：由车身框架中的加速度计测量的特定加速度。
- R_b^i : rotation from body frame to inertial frame.
 - 中文： R_b^i ：从本体坐标系到惯性坐标系的旋转。
- R_i^e : rotation from inertial frame to Earth-fixed frame.
 - 中文： R_i^e ：从惯性系旋转在地固系。
- $\mathbf{g}^e$: gravity acceleration vector.
 - 中文： $\mathbf{g}^e$ ：重力加速度矢量。

- Ω_{ie}^e : skew-symmetric matrix of Earth rotation rate.
 - 中文: Ω_{ie}^e : 地球自转速率的斜对称矩阵。
- $-2\Omega_{ie}^e \dot{\mathbf{x}}^e$: Coriolis acceleration.
 - 中文: $-2\Omega_{ie}^e \dot{\mathbf{x}}^e$: 科里奥利加速度。
- $-\Omega_{ie}^e \Omega_{ie}^e \mathbf{x}^e$: centrifugal term, often grouped with gravity in PPT. [Pos12 p22-p24]
 - 中文: $-\Omega_{ie}^e \Omega_{ie}^e \mathbf{x}^e$: 离心术语, 在 PPT 中通常与重力归为一组。 [Pos12 p22-p24]

3.8 LBS: path optimization by Dijkstra algorithm

中文: 3.8 LBS: Dijkstra算法的路径优化

Dijkstra algorithm:

中文: Dijkstra算法:

- Origin node: starting point.
 - 中文: 原点: 起点。
- Final node: destination.
 - 中文: 最终节点: 目的地。
- Based on iterative exploration of nodes. [Pos15 p19]
 - 中文: 基于节点的迭代探索。 [Pos15 p19]

Definitions:

中文: 定义:

- Estimated node: a node whose cost from origin has been estimated at least once.
 - 中文: 估计节点: 从原点开始的成本至少被估计过一次的节点。
- Explored node: a node that has already been explored.
 - 中文: 已探索节点: 已经被探索过的节点。
- Current explored node (CE): the node currently expanded. [Pos15 p19]
 - 中文: 当前探索的节点 (CE): 当前扩展的节点。 [Pos15 p19]

Algorithm:

中文: 算法:

1. Zero iteration: CE is the origin. Estimate costs from CE to all connected nodes.
中文: 1、零迭代: CE为原点。估计从 CE 到所有连接节点的成本。
2. In each following iteration, choose CE among already estimated nodes: the nearest/least expensive node to origin that is not yet explored.
中文: 2. 在接下来的每次迭代中, 在已估计的节点中选择 CE: 距离尚未探索的原点最近/最便宜的节点。

3. For each connected node, compute $\text{cost}(\text{CE}) + \text{cost}(\text{CE to connected node})$.
中文：3. 对于每个连接的节点，计算成本(CE) + 成本(CE 到连接的节点)。
4. Updated node cost is the minimum between the old estimate and new estimate.
中文：4. 更新后的节点成本是旧估计和新估计之间的最小值。
5. Store minimum-cost path to each estimated node and IDs of explored nodes.
中文：5. 存储到每个估计节点的最小成本路径以及探索的节点的ID。
6. Stop when destination has been reached, i.e. destination = CE. [Pos15 p19-p20]
中文：6. 到达目的地后停止，即目的地 = CE。 [Pos15 p19-p20]

Stored data:

中文：存储的数据：

- node;
 - 中文：节点;
- explored yes/no;
 - 中文：探索是/否;
- estimated yes/no;
 - 中文：估计是/否;
- minimum cost;
 - 中文：最低成本;
- path. [Pos15 p20]
 - 中文：小路。 [Pos15 p20]

4. Real oral exam expansion list from seniors

Equator / latitude / longitude definition

中文：赤道/纬度/经度定义

Use the geodetic coordinates answer:

中文：使用大地坐标答案：

- latitude: angle between ellipsoid normal through P and equatorial plane;
 - 中文：纬度：通过 P 的椭球法线与赤道面之间的角度;
- longitude: angle between meridian plane through P and origin meridian plane;
 - 中文：经度：经过 P 的子午面与原点子午面之间的角度;
- equatorial plane is the plane used as reference for latitude. [Pos03 p56]
 - 中文：赤道面是用作纬度参考的平面。 [Pos03 p56]

Ionospheric formula

中文：电离层公式

Say:

中文：说：

$$I_R^S = \pm \frac{A}{f^2} TEC_R^S$$

Positive for codes, negative for carrier phase. Parameters: A constant, f carrier frequency, TEC total electron content along path. [Pos08 p5-p7]

中文：正为代码，负为载波相位。参数： A 常数、 f 载波频率、 TEC 沿路径的总电子含量。
[Pos08 p5-p7]

Geodetic CRS

中文：大地测量CRS

Answer with: reference frame + ellipsoid + geodetic coordinates (φ, λ, h) . Latitude and longitude are angular coordinates defined with respect to ellipsoid normal, equatorial plane, and meridian planes; height is distance to ellipsoid. [Pos03 p56-p57]

中文：答案为：参考系+椭球体+大地坐标 (φ, λ, h) 。纬度和经度是相对于椭球法线、赤道面和子午面定义的角坐标；高程是到椭球体的距离。 [Pos03 p56-p57]

INS components, accelerometer, gyroscope, navigation equations

中文：INS 组件、加速度计、陀螺仪、导航方程

Use section 1.4 plus section 3.7. Teacher likely asks:

中文：使用第 1.4 节和第 3.7 节。老师可能会问：

- What are the 6 IMU outputs?
 - 中文：IMU 的 6 个输出是什么？
- What does accelerometer measure?
 - 中文：加速度计测量什么？
- What does gyroscope measure?
 - 中文：陀螺仪测量什么？
- How do you go from body frame observations to Earth-fixed position?
 - 中文：如何从载体坐标系观测到地固位置？

SPP formula and 5 bias sources

中文：SPP 公式和 5 个偏差源

Use SPP code equation in section 1.3.

中文：使用 1.3 节中的 SPP 代码方程。

Five sources:

中文：五个来源：

1. satellite orbit/clock/hardware bias;
中文：1. 卫星轨道/时钟/硬件偏差；
2. ionospheric residual;
中文：2. 电离层残余；
3. tropospheric residual;
中文：3. 对流层残余；
4. multipath;
中文：4. 多路径；
5. code noise/data quality. [Pos09 p21-p24]
中文：5. 代码噪声/数据质量。 [Pos09 p21-p24]

Multipath effect

- caused by reflection of GNSS signal before antenna;
 - 中文：由天线前GNSS信号反射引起；
- typical in urban scenarios;
 - complicated mathematical modeling;
 - 中文：复杂的数学建模；
- reaches some cm for phases and several m for codes;
 - 中文：相位达到几厘米，代码达到几米；
- periodic behavior around 20 minutes. [Pos08 p21]
 - 中文：20 分钟左右的周期性行为。 [Pos08 p21]

Near Earth orbit definition

中文：近地轨道定义

The POS PPTs discuss GNSS orbits and Keplerian/broadcast ephemerides, not a dedicated LEO definition in the extracted teacher slides. Safe answer if asked:

- Say that in these PPTs GNSS satellites are treated through Keplerian/quasi-Keplerian orbit parameters and ephemerides.

GPS detailed definition

中文：GPS详细定义

GPS is the USA GNSS constellation. It provides navigation data and user observables, mainly pseudoranges and phase observations, through signals containing carrier phases, binary codes and navigation message. [Pos05 p5, Pos07 p1-p2]

中文：GPS 是美国 GNSS 星座。它通过包含载波相位、二进制代码和导航消息的信号提供导航数据和用户观测值，主要是伪距和相位观测值。 [Pos05 p5, Pos07 p1-p2]

GPS / BeiDou / Galileo / GLONASS differences

Doppler

中文：多普勒

Doppler effect is defined as the phase shift on the received signal used to determine the relative velocity of the receiver with respect to the transmitter. [Pos14 p8]

中文：室内 PPT 将多普勒效应定义为接收信号的相移，用于确定接收机相对于发射器的相对速度。 [Pos14 p8]

5. Formula fast checklist

Distance:

中文：距离：

$$d = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2 + (Z_2 - Z_1)^2}, \quad d = cT$$

Ellipsoid:

中文：椭球：

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{b^2} = 1, \quad f = \frac{a - b}{a}, \quad e^2 = \frac{a^2 - b^2}{a^2}$$

Geodetic to Cartesian:

中文：大地测量到笛卡尔：

$$X = (R_N + h) \cos \varphi \cos \lambda, \quad Y = (R_N + h) \cos \varphi \sin \lambda, \quad Z = [R_N(1 - e^2) + h] \sin \varphi$$

$$R_N = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}$$

Height:

中文：高程：

$$h = H + N, \quad H = h - N$$

Code observation:

中文：码观测：

$$P=\rho+c(dt_R+d_R-d_t^S-d^S)+T+I+\varepsilon$$

Phase observation:

中文：相位观察：

$$L=\rho+c(dt_R+b_R-dt^S-b^S)+T-I+\lambda N+\varepsilon$$

Ionofree:

中文：无电离层组合：

$$P_3=\frac{f_1^2}{f_1^2-f_2^2}P_1-\frac{f_2^2}{f_1^2-f_2^2}P_2$$

Ionosphere:

中文：电离层：

$$I=\pm\frac{A}{f^2}TEC$$

PPP troposphere:

中文：PPP 对流层：

$$T=T_0+\delta T_Zm(el)$$

RSSI dB:

中文：接收信号强度指示分贝：

$$dB=10\log_{10}(P/P_0)$$

RSSI attenuation:

中文：RSSI 衰减：

$$P_R=kP_T\frac{G_TG_R}{4\pi d_{TR}^p}$$

ToA:

中文：到达时间：

$$ToA=\tau+\delta_R-\delta_T$$

Centroid:

中文：质心：

$$\mathbf{x}_u=\sum_iw_i\mathbf{x}_i,\quad\sum_iw_i=1$$

Least Squares:

中文：最小二乘法：

$$\hat{x}=(A^TQ^{-1}A)^{-1}A^TQ^{-1}(y-b)$$

Kalman:

中文：Kalman：

$$x_{t+1|t} = Tx_{t|t}, \quad \eta = y - Ax_{t+1|t}, \quad x_{t+1|t+1} = x_{t+1|t} + G\eta$$

Covariance propagation:

中文：协方差传播：

$$C_{new} = RC_{old}R^T$$

INS Earth-fixed:

中文：INS 地固：

$$\ddot{x}^e = f^e + g^e - 2\Omega_{ie}^e \dot{x}^e - \Omega_{ie}^e \Omega_{ie}^e x^e$$